

TOPAZ: Noise-Aware Quantum Circuit Architectures via Geometric Wells in the Parameter Landscape

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(Dated: 4 September 2026)

Near-term quantum circuits face a fundamental tension: reducing gate count limits decoherence, but too few gates fail to realise the target unitary. We introduce TOPAZ (Topology OPTimization Ansatz with Zoning), a classical framework that resolves this by treating circuit topology—the choice of which entangling edges are present—as a continuous design variable. Inspired by structural topology optimization, TOPAZ relaxes the unitarity constraint via polar decomposition, optimizes gate intensities and timings with a dual-MMA solver, and prunes low-intensity edges below a threshold. This yields circuits that are both shallower and, on simulated noise models, more noise-resilient, settling into what we call *structural wells*—noise-robust basins whose existence we characterize empirically across seeds, noise models, and a live device. On the 8-qubit FakeFez simulator, TOPAZ’s pruning step compiles its ansatz 34% shallower—relative to its own unpruned circuit—at negligible fidelity cost ($\Delta_{DM} = -0.0065$, the pruning-induced change in noise-aware density-matrix fidelity). On the 156-qubit ibm_fez processor, an 8-qubit experiment achieves a target-echo (inversion test) of $28.83\% \pm 3.94\%$ averaged over three same-day sessions: a $2.9\times$ improvement over the MMA-optimized product-Trotter baseline under matched parameter optimization, and $6.0\times$ over the untuned baseline. Noise-free evaluation of the identical circuits localizes this hardware gain to representation—TOPAZ stores $3.2\times$ more of the ideal target fidelity than the optimized baseline and $7.5\times$ more than the untuned one. The structural well transfers across synthetic noise models (efficiency $\eta = 0.9943$), and in simulation TOPAZ provides noise resilience complementary to post-processing methods such as ZNE and CDR, whose reliability the shallow topology preserves under the noisy VQE benchmark.

Keywords: TOPAZ, topology optimization, quantum error mitigation, noise-aware circuits, parametric Trotter expression, structural well

1. Introduction

A quantum computation is a sequence of gates, and on near-term hardware every gate is an opportunity for error. Depolarizing and damping channels erode the state, readout corrupts the output statistics, and coherence windows bound the total circuit time; variational algorithms must therefore execute complex operations within strict depth budgets [1, 2]. The conventional response is not to redesign the circuit but to keep it and edit the measurement statistics afterwards. Zero-noise extrapolation [3], Clifford data regression [4], and virtual distillation [5] all act after the gates have run—they stretch, fit, or distill. None of them alters which entangling edges are present in the compiled circuit, and a near-zero intensity that nobody deleted still becomes a physical CNOT.

We approach the problem from the other direction: design the circuit before it executes rather than correcting its output afterwards. TOPAZ relaxes the unitarity constraint during training, lets the optimizer drive unused entangling edges down to zero intensity, and prunes them before compilation. The engine is the *local parametric Trotter expression* (LPTE): an N -qubit linear chain in which each nearest-neighbour pair $(i, i+1)$ carries a sum of three Pauli-rotation terms

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-iP \delta \tau_{i,P}}, \quad (1)$$

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\max}]$. The normalization $\sum_P r_{i,P} = 1$ is enforced via softmax: the optimizer works with unconstrained weights $\tilde{\rho}$ and sets

$r_{i,P} = \exp(\tilde{\rho}_P) / \sum_{P'} \exp(\tilde{\rho}_{P'})$, ensuring the simplex constraint without projection. Each block is a non-unitary operator; it is projected to its nearest unitary by the polar factor

$$\mathcal{P}_U(A) = A (A^\dagger A)^{-1/2}, \quad (2)$$

and the circuit is the product of the per-pair unitaries. Because the relaxation is continuous, gradients can flow while an edge is being removed, something a strictly unitary ansatz cannot do.

This has the shape of a classical structural topology-optimization problem [6, 7]: a discretized solid, a density variable on every element, a compliance residual, and an optimizer that removes material the residual does not need. TOPAZ draws direction from that field. The density on edge p is the total intensity $\kappa(p) = \sum_P r_{p,P}$; the residual is the fidelity under a Kraus noise model; and the optimizer is the method of moving asymptotes, applied twice—once for intensities ρ , once for timings τ . Edges whose intensity falls below a registered threshold $\tau_\kappa = 0.05$ are pruned, as low-density elements are removed in the solid isotropic material with penalization (SIMP) method of structural optimization [6]. We emphasize that the debt to SIMP is one of inspiration, not of method: TOPAZ prunes by an explicit threshold and applies neither SIMP’s power-law penalization of intermediate densities nor its compliance objective. The continuous-relaxation-and-prune strategy is the common thread; the machinery is TOPAZ’s own.

TOPAZ has three mechanisms. (1) The LPTE sum with polar retraction provides a continuous relaxation of the unitarity constraint, allowing gradients to flow around barriers that would

block a strictly unitary optimizer. (2) Parameter zoning treats ρ and τ as heterogeneous design variables updated by separate dual-MMA solvers with independent asymptotes and move limits; a discrete stage then removes low-intensity entangling edges. (3) The optimizer consistently reaches a *structural well*—a basin in the noise-deformed parameter landscape where the noisy output is robust and, empirically, stable across seeds, noise models, and hardware.

We demonstrate the framework on the 8-qubit FakeFey simulator and, in an 8-qubit experiment on the quietest path of the 156-qubit `ibm_fez` (Heron r2) processor. Relative to its own unpruned ansatz, TOPAZ’s pruning step makes meshes 34% shallower (internal reduction) at negligible fidelity cost from the depth reduction alone ($\Delta\text{DM} = -0.0065$, the pruning-induced change in noise-aware density-matrix fidelity); post-processing ZNE on top of TOPAZ adds a further +0.0667 DM gain (Table VIII), for a combined improvement that is not merely additive. On hardware, the submitted circuit uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline, and its noiseless one-way state-preparation fidelity is 56.62%. The measured target-echo (inversion test), averaged over three same-day sessions, is $28.83\% \pm 3.94\%$ versus $4.78\% \pm 0.34\%$ for the untuned product-Trotter baseline ($6.0\times$, $z \approx 22$) and $10.08\% \pm 0.48\%$ for the MMA-optimized baseline ($2.9\times$, $z \approx 17$). Noise-free evaluation of the identical circuits isolates the origin of this gain: TOPAZ stores $3.2\times$ more of the ideal target fidelity than the MMA-optimized baseline (0.5662 vs. 0.1768) and $7.5\times$ more than the untuned one (0.0758), so the hardware advantage is representational, not a post-processing artefact—pre-circuit topology optimization succeeds where extrapolation-based post-processing is inconsistent. All ZNE, CDR, and VD results reported in this paper are computed from our own implementations applied to the same simulated noise models used for training.

The contributions of this work are: (i) a continuous operator relaxation for quantum circuits via polar retraction, with the radial (stretching) mode of the polar factor observed to leave the unitary rotation invariant (Observation 1) and the unital noise baseline observed to leave critical points invariant (Observation 2); (ii) an end-to-end synthesis pipeline combining LPTE with dual-MMA and automated pruning, yielding a 34% depth reduction relative to its own unpruned ansatz at $\Delta\text{DM} = -0.0065$; (iii) a tensor-dot Kraus contraction engine running at $O(2^{2N})$ per term, a $3,838\times$ wall-clock speedup over superoperator methods at $N = 10$; and (iv) a demonstration of a target-echo (inversion test) of $28.83\% \pm 3.94\%$ (mean of three same-day sessions) in an 8-qubit experiment on the 156-qubit Heron r2 processor, a $2.9\times$ improvement over the MMA-optimized product-Trotter baseline under matched parameter optimization ($6.0\times$ over the untuned baseline).

2. Continuous relaxation and the structural well

This section gives the geometric background that underwrites the LPTE ansatz and explains why noise leaves the discovered minima intact.

2.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \mathbb{C}^{d \times d} : U^\dagger U = I\}$, a compact Lie group of dimension d^2 with $d = 2^N$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \mathbb{C}^{d \times d}$ without the unitarity constraint, so gradients can flow around barriers that would block a strictly unitary optimizer. Every invertible $A \in \text{GL}(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [8, Sec. 8.1]; the polar factor Q is the closest unitary to A in the Frobenius norm. Projecting back to $U(d)$ —*unitary regularization*—pins singular values to unity, smoothing the loss landscape and stabilizing gradients. In LPTE the per-pair block $B_{(i,i+1)}$ is lower-bounded by the clipping floor $\max(\rho, 10^{-6})$, so $\sigma_{\min}(B)$ stays positive and the polar factor is unique, deterministic, and differentiable on the active block.

2.2 Expressibility through dynamical Lie algebras

The set of reachable states of an ansatz is determined by the Dynamical Lie Algebra (DLA) of its generators [9]. Without entanglement the generators are local Pauli operators that commute across qubit boundaries, and the Lie closure has dimension only $3N$, which severely restricts the optimizer. An entangling generator $X_1 X_2$ breaks this: cross-pair commutators such as $[X_1 X_2, Y_2 Y_3] = 2i X_1 Z_2 Y_3$ generate new non-local strings and enlarge the reachable manifold. A generic Hardware-Efficient Ansatz (HEA) mixes local and arbitrary non-local generators, so its DLA closes rapidly to $\mathfrak{su}(2^N)$ (dimension $4^N - 1$), a growth associated with barren plateaus [10]. By contrast, the LPTE ansatz restricts each block to an Abelian $\{XX, YY, ZZ\}$ triple, and chained blocks grow the manifold through cross-pair commutators without closing to $\mathfrak{su}(2^N)$ at the depths used here. This block-structured growth is what makes local pruning safe: removing one chain removes only its own block algebra, leaving the reachable manifold of the remaining blocks unchanged. We do not claim a barren-plateau theorem from this picture; the canyon-river Hessian measurements (§7.7.1) are consistency checks, not proofs.

2.3 The structural well and topological invariance

Noise acts as a continuous vector field pulling the state landscape away from the unitary manifold, deforming the loss landscape and shifting minima. Instead of correcting the output after the fact, TOPAZ searches for a *structural well*—a region of the parameter manifold where the noisy output is intrinsically robust (Fig. 1). We formalize this with a precise terminology (Definition 1), a small set of standard facts stated

as observations, one explicit assumption, and one conjecture; the cross-noise evidence is empirical. The definition pins down what object we mean by a well; its geometric characteristics—curvature bands, effective rank, basin volume proxy—are then *measured* properties of that object, reported as data in the canyon-river analysis (§7.1), not as theorems. This separation is deliberate: the formal object is the sublevel-set basin of Definition 1, and everything asserted about its quantitative shape is an empirical characterization.

Definition 1 (Parameter space). The optimization domain is $\mathcal{X} = [0, 1]^M \times [0, \tau_{\max}]^M$, where M is the number of entangling pairs. This space is closed and bounded, hence compact. The loss $L : \mathcal{X} \rightarrow \mathbb{R}$, $L(\rho, \tau) = 1 - |\langle \psi_{\text{ideal}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$, is continuous. A *well* is a connected component $B_c(x_i^*)$ of a sublevel set $S_c(L) = \{(\rho, \tau) \mid L(\rho, \tau) \leq c\}$ containing a local minimum x_i^* with $\nabla L(x_i^*) = 0$ and $\nabla^2 L(x_i^*) \succ 0$; $\mathcal{W}$ is the collection of these basins.

Remark 1 (Wells observed, not derived). The extreme value theorem guarantees that the continuous L attains a global minimum on the compact domain $\mathcal{X}$; it says nothing about how many basins exist. That the landscape hosts multiple wells is an experimental observation (different seeds converge to different floors). Different seeds land in different wells because each freezes a different sparse subset of parameters.

The wells are not artefacts of a single run. Across ten independent seeds of the dual-MMA optimizer, TOPAZ repeatedly settles into the same regional floor: mean DM 0.737 ± 0.066 , with seed-to-seed floor variation $\Delta F < 0.05$, indicating that the discovered basin is reproducible rather than a fragile single-trajectory attractor. The geometry of the basin is consistent with that picture. The Hessian of the loss at the converged point has an effective rank $r_{\text{eff}} \approx 3$ —on a ~ 14 -parameter problem, 34 of 42 eigen-directions are flat—so the well is a low-dimensional valley embedded in larger plateau structure, with a vanishing smallest eigenvalue $\lambda_{\min}(\nabla^2 L) = 0$ exactly at the basin boundary, marking the transition out of the well. The parameter stiffness is strongly heterogeneous: ρ -directions are uniformly stiff, each carrying curvature ~ 870 ($\kappa_\rho \approx 1$ in the normalized units of §7.1), while τ -directions have a few extremely stiff directions amid many soft ones, $\kappa_\tau \approx 9 \times 10^3$ combined-stiffness ratios ranging over two orders of magnitude, with optimal timings spread non-uniformly across $\tau \in [0.004, 35]$. Each seed freezes a different sparse subset of entangling pairs (e.g. seed 3 removes three of seven pairs), yet all settle on the same floor—evidence that the well is a connected family of sparse subspaces with consistent geometry rather than a single isolated minimum.

The two observations and two conjectures below govern, respectively, the classical optimization path and the effect of noise on it. They are not presented as theorems; the guarantees of this work are established empirically in the results sections.

Observation 1 (Radial perturbations do not rotate the polar factor). Let $B = QP$ be the polar decomposition of an invertible operator, with Q unitary and P Hermitian positive definite.

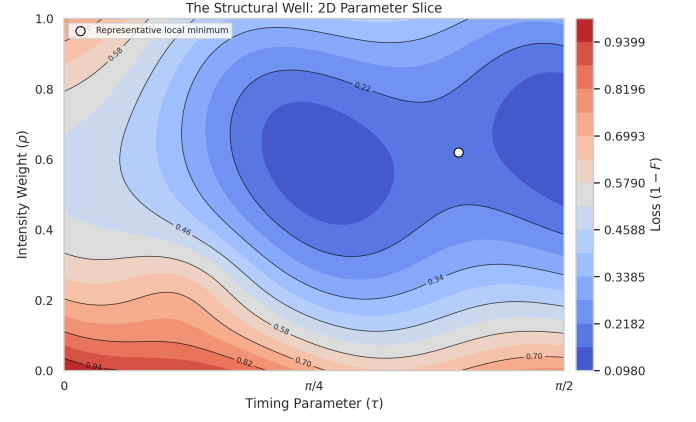


FIG. 1 The structural well on FakeFез, projected onto a 2D (ρ, τ) slice. Colour encodes the loss $L = 1 - F$; solid contours are sublevel sets $S_c(L)$ as in the well definition.

Consider the scaling (radial) perturbation $\delta B = \varepsilon B$. Then $B + \delta B = (1 + \varepsilon)B = Q(1 + \varepsilon)P$ is already a polar decomposition, since Q is unitary and $(1 + \varepsilon)P$ is Hermitian positive definite for $1 + \varepsilon > 0$, so by uniqueness of the polar factor the rotation vanishes: $\delta Q = 0$.

Remark 2 (General perturbations do rotate Q). For a general perturbation δB , the polar factor does rotate. Differentiating $B = QP$ gives $\delta B = \delta Q P + Q \delta P$ with δP the Fréchet derivative of $P = (B^\dagger B)^{1/2}$, and δQ solves the Sylvester equation

$$\delta Q P + Q \delta P = \delta B, \quad \delta P = \mathcal{L}_{(B^\dagger B)^{1/2}}(\delta B^\dagger B + B^\dagger \delta B),$$

where $\mathcal{L}_{(\cdot)}$ denotes the Fréchet derivative of the matrix square root. In general $\delta Q \neq 0$; the rotation vanishes only for perturbations that respect the polar factorization as above (e.g. the radial case of Observation 1). The unitary regularizer of Eq. (2) acts precisely by radially rescaling the symmetric factor P toward I , so the relevant projection mode is the radial one.

Observation 2 (Unital noise is an affine baseline). A standard fact about completely positive trace-preserving (CPTP) maps is that any such map splits into a unital part $\mathcal{E}_{\text{uni}}$ and a non-unital part [35, 36]. We restate the unital part here because it is the precise reason the optimizer’s critical-point structure survives under the dominant noise component. For the dominant unital part (e.g. depolarizing noise) the noisy output is affine in the ideal state,

$$\sigma_{\text{noisy}}^{(\text{uni})} = (1 - p) \sigma_{\text{ideal}} + \frac{p}{d} I,$$

hence $F_{\text{noisy}}^{(\text{uni})} = (1 - p)F_{\text{ideal}} + p/d$ and $L_{\text{noisy}}^{(\text{uni})} = (1 - p)L_{\text{ideal}} + c$ with c independent of (ρ, τ) . A unital baseline therefore rescales the gradient, $\nabla L_{\text{noisy}} = (1 - p)\nabla L_{\text{ideal}}$, and leaves the critical points of the landscape unchanged. This is a known result; we state it not as a contribution but because it justifies treating the landscape’s critical-point structure as approximately noise-invariant under the unital component, which is what the empirical approach relies on.

Assumption 1 (Suppression of non-unital deformation). Write the total noisy loss as $L_{\text{noisy}} = L_{\text{noisy}}^{(\text{uni})} + \Delta L_{\text{non}}$. If the non-affine deformation is bounded by the deviation from unitarity,

$$\Delta L_{\text{non}} \leq K' \|B^\dagger B - I\|_F,$$

with K' independent of (ρ, τ) , then, because the regularizer (Eq. 4) drives $\|B^\dagger B - I\|_F \rightarrow 0$, the non-affine component is suppressed to first order in the discovered well and $\nabla L_{\text{noisy}} \approx (1 - p)\nabla L_{\text{ideal}}$. We state this as an assumption rather than a theorem: the regularizer penalizes, but does not by itself guarantee, convergence to unitarity, and the bound on ΔL_{non} is a modelling hypothesis. Its numerical support comes from two complementary places. (i) The marginal evidence is the polar-projection ablation of Table I: removing the polar projection leaves $\|B^\dagger B - I\|_F$ large and halves the DM fidelity (0.334 vs. 0.737), exactly the direction the bound predicts—large deviation from unitarity, large non-affine loss. (ii) At the converged point the residual deviation is small in absolute terms: the regularizer penalty $R = \|B^\dagger B - I\|_F$ converges to 0.0252 on the quietest-path product PTE, and the noisy fidelity sits within 2.2% of the ideal (0.6329 vs. 0.6545), consistent with a small non-affine correction riding on an approximately affine unital baseline. A trajectory-level scatter plot of ΔL_{non} versus $\|B^\dagger B - I\|_F$ is not claimed here; the two endpoints above bracket the regime the assumption is used in, and the transfer matrix (§6.4) tests it end-to-end under all six noise models.

Conjecture 1 (Well invariance under unital-dominant noise). Within the regime of Assumption 1, the ideal and noisy sublevel sets are related by a monotone reparametrisation of the threshold, $S_c(L_{\text{noisy}}) = S_{\tilde{c}}(L_{\text{ideal}})$ with $\tilde{c}$ affine in c . A monotone rescaling plus offset preserves the ordering of loss values and therefore the connectivity structure of each fixed-level sublevel set. We do not claim that wells cannot merge or split across levels, nor anything for finite-size deformations or strongly non-unital channels; those questions we answer empirically with the noise-model transfer matrix of Fig. 5.

The guarantees above are local: TOPAZ operates on a constrained subspace—the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective—so the smoothness claims are scoped to the local regularity of the structural well, not to the global trap-free theorem of full controllability [11].

3. The Parametric Trotter Expression family

Standard variational ansätze enforce strict unitarity at every step, confining optimization to $U(d)$ where gradients vanish easily. The Parametric Trotter Expression (PTE) [12] relaxes this by letting intermediate operators leave the unitary manifold and projecting back. Three formulations span the locality spectrum.

Global PTE. The full sum $B = \sum_j \rho_j e^{-iH_j \delta \tau_j}$ is mathematically unconstrained but computationally expensive: each polar decomposition step requires an $O(d^3)$ matrix inversion

with $d = 2^N$, giving $O(2^{3N})$ per evaluation. Its 8-qubit inversion test registers 0.004, at the 1/256 noise floor, and the cost grows exponentially with qubit count. Global PTE was not used for the headline experiments.

Product PTE. Enforcing a term-by-term Trotter product, $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta \tau_j}$, restores hardware compilation. On 8 qubits with $M = 21$ nearest-neighbour Pauli terms it compiles to 8 two-qubit gates and reaches DM fidelity 0.6329 under FakeFez—a 9.35× gain over the random baseline (0.0677). But every factor is unitary by construction, so the product form cannot drive intensities to zero and topological pruning is impossible. *Gate-count audit.* Throughout, “two-qubit gate” denotes a native entangling gate (ECR on `ibm_fez`, CNOT on FakeFez) counted after transpilation; single-qubit rotations are excluded. All baseline and LPTE circuits in this work were transpiled with Qiskit `optimization_level=3` onto the device native-gate basis; the pre/post-transpilation entangling-gate counts for every circuit, together with the generated QASM, are recorded in the public data release (Table VI reports the post-transpilation counts). The reduction from 21 abstract Pauli terms to 8 native entangling gates—specific to the *product* form above—reflects term-merging and local-rotation absorption by the transpiler at this optimization level; the LPTE ansatz, by contrast, retains 21 two-qubit gates (Table VI) because its per-pair polar projection blocks the term-merging route (§3).

Definition 2 (Local Parametric Trotter Expression). The *local parametric Trotter expression* (LPTE) is the ansatz family $\{U_{\text{LPTE}}(\rho, \tau)\}$ defined as follows. On an N -qubit linear chain, each nearest-neighbour pair $p = (i, i+1)$ carries a block $B_{(i,i+1)}(\rho, \tau)$ of Eq. (1): a sum of three Pauli rotations with intensity $r_{i,p}$ and timing $\tau_{i,p}$. Each block is projected to its nearest unitary by the polar factor of Eq. (2), $U_{(i,i+1)} = \mathcal{P}_U(B_{(i,i+1)})$, and the circuit is the ordered product $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$. The entangling graph is the chain itself, and the intensity $\kappa(p) = \sum_P r_{p,P}$ is the design variable that TOPAZ drives to zero for pruning.

Remark 3 (Terminology). We use the term “Trotter” only loosely. The LPTE is a sum of Pauli rotations projected back to unitarity; it is not a Trotter–Suzuki decomposition, and no explicit time-slicing or commutator-bounded splitting is involved. The name signals only that each block factorizes into a product of local rotations, in the sense of the product-form expressions used throughout.

Local PTE (LPTE). LPTE localizes the non-unitary sum within nearest-neighbour pairs (Definition 2). On 8 qubits with 7 pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters—the same parameter count as product PTE but 2.6× the gate count. The difference that matters here is that the per-pair SVD projection lets the optimizer independently set each pair’s intensity: it can discover which pairs carry the noise-resilience behaviour and assign them nonzero intensity, while letting others fall to zero so that an idle edge can be pruned without rewriting the rest of the circuit. Product PTE’s unitary factors have no such selection mechanism.

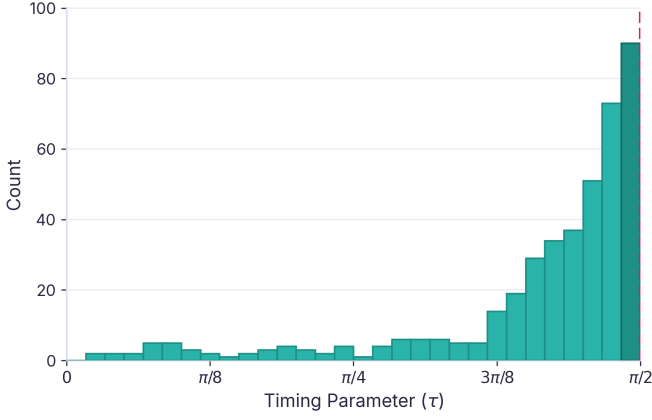


FIG. 2 Distribution of optimised timings τ across the LPTE benchmark seeds. Dashed line: $\tau = \pi/2$, where the block of Lemma 1 projects to a maximally entangling SWAP.

3.1 The $\pi/2$ timing cluster is a SWAP identity

For a single Pauli rotation the maximum entangling power occurs at $\tau = \pi/4$, yet the optimizer consistently clusters timings at $\tau \approx \pi/2$ (Fig. 2). The following lemma shows why this is benign.

Lemma 1 (The $\pi/2$ cluster is a SWAP identity). *For a local block $B(\tau) = e^{-i\tau XX} + e^{-i\tau YY} + e^{-i\tau ZZ}$, at $\tau = \pi/2$, $B = -i(XX + YY + ZZ)$. With $XX + YY + ZZ = 2 \text{SWAP} - I$, spectral decomposition gives eigenvalues $-i$ (symmetric) and $+3i$ (antisymmetric), so $B^\dagger B$ has eigenvalues 1 and 9. The polar projection $(B^\dagger B)^{-1/2}$ scales the symmetric subspace by 1 and the antisymmetric by $1/3$, yielding a U with eigenvalues $-i$ and $+i$: exactly $-i \text{SWAP}$. At $\tau = \pi/2$ the PTE sum plus projection deterministically produces a maximally entangling SWAP.*

3.2 Objective functions

The optimizer minimizes the noisy fidelity loss

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (3)$$

or, for a target state, the state fidelity $L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$. A regularizer keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (4)$$

3.3 Self-pruning under noise

LPTE self-prunes: during optimization the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters freeze. Across ten seeds under FakeFaz typically 8–24 of the 42 (ρ, τ) dimensions survive, corresponding to about 9 of 21 entangling pairs active. Table I shows that self-pruning requires the full stack (PTE ansatz, polar projection, dual-MMA).

TABLE I Full ablation on the 8-qubit FakeFaz benchmark (LPTE, 21 two-qubit gates). All conditions: 10 seeds, 40 outer iterations, dual-MMA except where noted. Significance versus full TOPAZ at a Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q	Active	d
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12^{ns}
Single MMA	0.623 ± 0.166	19.9	19.9	0.80^{ns}
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02^{ns}
No polar decomp.	0.334 ± 0.170	16.7	15.8	2.14^{***}
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97^{***}
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09^{***}
Random PTE	0.056 ± 0.031	21.0	21.0	8.69^{***}

*** $p < 0.001$ (Bonferroni-passing). Cohen's d vs. full TOPAZ; ns, not significant.

Exact state preparation under the same noise scores 0.924 ± 0.003 with all 21 gates active.

Full TOPAZ (DM 0.737 ± 0.066) beats product-only PTE (0.279 ± 0.175 , $d = 3.09$), HEA (0.329 ± 0.110 , $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $d = 2.14$). Removing the polar projection halves the fidelity, confirming that it is central to the gain.

Parametric efficiency is summarized in Table II: the best seed of LPTE reaches 0.9085 with 42 parameters and 21 gates, whereas HEA uses ~ 96 parameters and 28 gates for 0.1517—a $6\times$ gap in fidelity per parameter. (These are best-seed values; the multi-seed mean of full TOPAZ is 0.737 ± 0.066 , Table I.)

TABLE II Parameter efficiency of the PTE family versus HEA on 8 qubits under FakeFaz. DM Fidelity is the best-seed value (cf. the 10-seed mean 0.737 ± 0.066 in Table I); best-seed is reported here because these rows feed the single-trajectory comparison of Table VI.

Ansatz	Parameters	2Q	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

4. Methods: the TOPAZ engine

4.1 Tensor-dot Kraus contraction

Applying k -local Kraus operators to a pure state costs $O(2^N)$ per term. For a density matrix the naive construction of the $2^{2N} \times 2^{2N}$ superoperator costs $O(2^{3N})$. The tensor-dot engine reframes ρ as a tensor with $2N$ indices and contracts locally. Applying a Kraus channel $\rho \mapsto \sum_a K^{(a)} \rho K^{(a)\dagger}$ acts only on the active qubit set $\mathcal{S}$; the spectator set $\mathcal{S}^c$ passes through unchanged, so the channel is

$$\rho'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a \sum_{i_S, j_S} K_{i'_S, i_S}^{(a)} \rho_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger}, \quad (5)$$

where $\mathcal{S}^c$ denotes the spectator qubits, the input density matrix is $\rho_{i_S, i_{S^c}; j_S, j_{S^c}}$, and the output carries the contracted (primed) active indices i'_S, j'_S with the spectator indices i_{S^c}, j_{S^c} unchanged. Because only the $2|\mathcal{S}|$ active indices are contracted

while the $2N - 2|S|$ spectator legs are untouched, each term costs $O(2^{2|S|})$ rather than $O(2^{2N})$. This yields a $3,838\times$ wall-clock speedup over dense Kraus at $N = 10$ (Table III). The ratio of the per-term complexity classes alone, $O(2^{3N})/O(2^{2N}) = 2^N$, predicts $\sim 1024\times$ at $N = 10$; the additional $3.7\times$ beyond this exponent-ratio bound is accounted for by per-term constant factors and by the fact that the tensor-dot engine contracts only the $2|S|$ active indices of each term while the dense superoperator materializes the full 2^{3N} -dimensional channel. The measured speedup is therefore consistent with—rather than in tension with—the stated complexity.

TABLE III Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity/term
Qiskit dense Kraus	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^{2N})$
Speedup	$\sim 3838\times 2^N (= 2^{10}=1024\times \text{exp. bound})$	

The engine plays different roles on different paths: on the legacy triple-channel path it performs both ideal and noisy evolution; on the FakeFez path it computes only the noiseless target, delegating noisy evaluation to AerSimulator; on live hardware it has no role, since the backend executes gates directly.

4.2 Noise models

TOPAZ runs against two families of noise. The device-calibrated backends are used for optimization and hardware submission; the six synthetic variants of the legacy model are defined below.

Triple-channel correlated (legacy). Depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, and phase damping at $\lambda = 0.012$, with multi-qubit correlations decaying as $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$, $\xi = 2.0$ [13].

FakeFez. IBM’s calibrated snapshot of `ibm_fez` (Heron r2, 156 qubits) with per-gate error rates and qubit-specific T_1 , T_2 , and readout errors [14]. All headline experiments use FakeFez during optimization.

FakeMontrealV2. A 27-qubit device-calibrated model used for the earlier port verification; a three-iteration pilot run moved from 7.4% to 36.0% DM fidelity.

Live ibm_fez. The optimized (ρ, τ) from FakeFez are submitted to live Heron r2 [15] without re-optimization.

5. The TOPAZ optimization framework

5.1 Dual-MMA optimizer

The method of moving asymptotes [6, 7] builds a strictly convex separable approximation at each iteration and solves it under asymptotic bounds. *Dual-MMA* interleaves two such instances, one per parameter class, because ρ and τ differ by roughly four orders of magnitude in finite-difference step

size ($\epsilon_\rho = 10^{-4}$ versus $\epsilon_\tau = \pi/4$); a single shared move-limit budget misallocates updates. Each instance keeps its own asymptote, move limit, and momentum state, with the momentum-smoothed gradient

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4. \quad (6)$$

Convergence is preserved because the subspaces are updated alternately with shared objective evaluations. When a target algorithm carries an extra parameter class (e.g. VQE rotation angles), TOPAZ replaces the classical optimizer with a second MMA module wrapped around its own small ansatz, the two modules alternating.

5.2 Topology optimization by sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ whose edge p carries normalized intensity $\kappa(p) = \sum_P r_{p,P}$. The active pruning protocol has three stages: (1) solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on the full edge set with no artificial floor; (2) remove pairs with $\kappa(p) < \tau_\kappa = 0.05$, i.e. $E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}$; (3) re-optimize on the reduced domain from fresh restarts.

TABLE IV Multi-seed topology optimization ($\tau_\kappa = 0.05$, 5 seeds). F, full (Stage 1); S, sparse (Stage 3). Pruned is the number of entangling pairs removed.

Seed	DM fid.		2Q		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

Across five seeds (Table IV), TOPAZ’s own pruning step reduces gate count by 14% ($18.6 \rightarrow 16.0$) and schedule depth by 34% ($72.4 \rightarrow 48.0$) relative to its unpruned ansatz—an internal, within-pipeline reduction—at a mean DM loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while improving DM by $+0.0081$.

Lemma 2 (Topological sparsification). *If an entangling pair contributes only stretching (non-unitary) behaviour and not to the ideal rotation, its contribution to L_{prim} is deleted by the polar projection, so the primary gradient with respect to that pair vanishes. Because TOPAZ’s step size scales with the gradient, a zero gradient drives the step to zero and the parameter decays onto the clipping floor $\max(\rho, 10^{-6})$, where it freezes and leaves the active search. The regularizer in Eq. (4) only stabilises the active blocks; it does not itself produce the pruning.*

5.3 Algorithmic compatibility

Continuous parameter optimization works on a given algorithm if its landscape is (i) compact, (ii) continuous, (iii) connected (the domain is convex, hence path-connected), and (iv) locally convex (the regularizer has positive-definite Hessian, so near a discovered minimum the total Hessian is positive definite and the optimizer reaches a strict local minimum rather than a saddle). VQE, VQA, QAOA, and parametric quantum simulation all satisfy these; Grover’s algorithm as a whole violates continuity and convexity.

6. Results

6.1 Trajectories under correlated noise

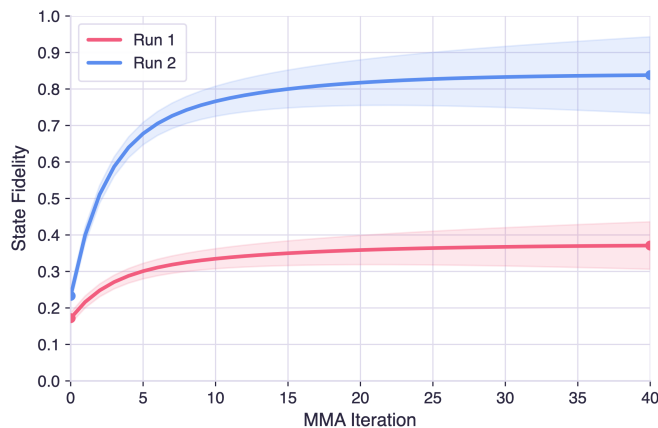


FIG. 3 Two representative 8-qubit LPTE trajectories under triple-channel correlated noise (Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838). Shaded bands are ± 1 standard deviation across the 10-seed ensemble.

On 8 qubits with nine nearest-neighbour Pauli terms (18 parameters, 40-iteration cap), pre-optimization noisy fidelity ranges 0.07–0.70; post-MMA it rises to 0.30–0.88, an absolute gain of 18–23 points (Fig. 3). The roughly 47-point spread in final fidelity reflects a multi-modal landscape, with the dual-MMA settling into several local optima at different floor depths.

6.2 Noise-model-aware VQE on H₂/H₄

In the main TOPAZ pipeline the optimizer uses a noise-model-aware simulator that evaluates both the noisy density-matrix fidelity and the ideal-state fidelity (as in the Loschmidt-echo objective of §6.5); it is this full objective that produces the headline 28.8% echo result on `ibm_fez`. The H₂/H₄ VQE benchmark deliberately restricts that regime: the optimizer is given only raw noisy energy measurements—no target state, no ideal observable, and no information about the post-processing it will later be stacked with. This restriction tests whether LPTE’s circuit topology still confers noise resilience even when the ideal-state constraint is removed—a harder regime that isolates the structural contribution of the ansatz from the optimizer’s noise-model advantage. Three ansätze—LPTE,

HEA (depth 3 for H₂, depth 5 for H₄), and HVA $d = 2$ —are compared on the 4-qubit H₂ path [136, 143, 142, 141] and the 8-qubit H₄ path [125, 124, 123, 136, 143, 142, 141, 140] under FakeFez (Table V). All runs use ZNE, CDR, and VD post-processing across 10 seeds (except HVA, for which ZNE+CDR was not run; Table V); the ZNE, CDR, and VD implementations are our own, applied to the same simulated noise model used for training.

TABLE V Combined H₂/H₄ VQE benchmark under FakeFez. All baselines were implemented with the same transpiler, backend snapshot, shot budget, and optimizer budget as TOPAZ.

Mol.	Ansatz	Noisy ΔE_{FCI} (mHa)	Noiseless (mHa)	ZNE+CDR (mHa)	Seeds	Conv.
H ₂	LPTE	71.96 $\pm$ 16.8	26.72 $\pm$ 14.5	26.16 $\pm$ 14.8	10	100%
H ₂	HEA ($d=3$)	54.04 $\pm$ 10.5	24.23 $\pm$ 8.8	0.0*	10	30%
H ₂	HVA ($d = 2$)	748.28 $\pm$ 25.3	492.01 $\pm$ 162.8	†	10	0%
H ₄	LPTE	235.15 $\pm$ 77.4	83.82 $\pm$ 74.5	83.75 $\pm$ 74.4	10	100%
H ₄	HEA ($d=5$)	889.90 $\pm$ 232.8	710.35 $\pm$ 241.0	713.16 $\pm$ 243.3	10	0%
H ₄	HVA ($d = 2$, 368p)	2720.55 $\pm$ 6.5	2270.06 $\pm$ 462.7	†	10	0%

*Clamped to FCI (see text). †ZNE+CDR not run on HVA. HVA runs were limited to 500 optimizer iterations ($d=2$, 28–368 parameters) and converged on no seed, so their error bars should be read as estimates of a non-converged baseline.

On H₂, LPTE plus ZNE+CDR reduces the mean energy error from the 71.96 mHa noisy baseline to 26.16 mHa—a mean 2.75 $\times$ reduction, and up to 8.96 mHa ($\approx$ 5.6 kcal/mol) on the best seed—though still above the 1.6 mHa (1 kcal/mol) chemical-accuracy bar. On H₄, LPTE’s ZNE+CDR (83.75 mHa) recovers the noiseless floor (83.82 mHa) almost exactly, because the shallow depth ($\sim$ 8 effective two-qubit gates) preserves the noise-energy relationship that extrapolation requires. HEA extrapolations violated physical bounds on 7 of 10 seeds and required clamping; LPTE required none.

We note explicitly that in the unmitigated regime LPTE does *not* win on H₂: its raw noisy error is worse than HEA’s (71.96 vs. 54.04 mHa). However, on H₄, LPTE’s raw error (235.15 mHa) is dramatically lower than both HEA (889.90 mHa) and HVA (2720.55 mHa), because the deeper baselines fail to converge to competitive solutions on 8 qubits (both report 0% convergence; Table V). The advantage that Table V shows across both molecules is therefore one of *mitigation-compatibility*—LPTE’s shallow topology keeps the noise-energy relationship intact enough that correction techniques remain reliable on top of it (and remain physically valid, requiring no clamping), whereas the deeper baselines degrade under the same pipeline. This is a narrower claim than unmitigated noise-resilience for H₂, and we state it as such: the VQE results establish that TOPAZ/LPTE composes well with post-processing on this restricted (raw-energy-only) regime, not that its topology is intrinsically more noise-robust than competing ansätze in the absence of mitigation.

6.3 Cross-Hamiltonian transfer

To test whether TOPAZ reduces to a generic reusable circuit, an H₄-optimized topology (Layout A, 263 entangling gates) was re-entered as the ansatz for an 8-qubit TFIM at the critical point $J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Holding the structure

fixed and re-optimizing only the angles gives noisy $\Delta E_{\text{FCI}} = 1211.0$ mHa; running the full pipeline from scratch gives 932.1 mHa. The transferred topology is worse by 278.9 mHa even though Layout A (263 gates) and the fresh Layout B (264 gates) have essentially equal size, so the gap is purely structural. TOPAZ optimizes for the target’s entangling structure; it does not reduce to a reusable circuit.

6.4 Comparison on FakeFez and the noise-model transfer matrix

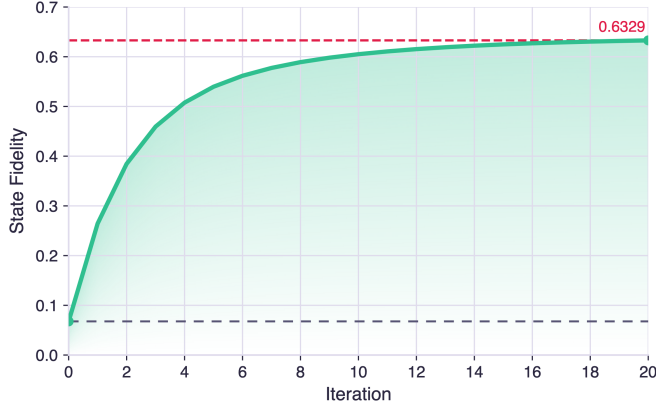


FIG. 4 FakeFez product-PTE convergence (8 qubits, 42 parameters). The trajectory descends over the first 5–10 iterations and flattens at the structural-well floor; dashed line marks $F = 0.6329$.

TABLE VI Comparison on FakeFez (quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion-test fidelity while using roughly half the gates. LPTE DM/Inv./CDR/VD are best-seed values of the trajectory in Fig. 4; the 10-seed mean after pruning is 0.737 ± 0.066 (Table I).

Method	DM	Inv.	CDR	VD	2Q	Depth
Exact prep (raw)	0.8501	0.7150	0.2313	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 lyr.)	0.1517	0.1305	—	—	28	57
TOPAZ product	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda = 0$, not measured. CDR and VD are single-method results applied to the LPTE circuit; they were not run stacked with ZNE. All mitigation methods (ZNE, CDR, VD) are implemented in-house and run against our own density-matrix simulations.

On the FakeFez simulator, the full TOPAZ LPTE ansatz (21 two-qubit gates, depth 97; Table VI) reaches ideal fidelity 0.9829 (approximation loss 1.71%) and, at its best seed, 0.9085 under noise (noise loss 7.41%; the 10-seed mean after pruning is 0.737 ± 0.066); exact preparation loses 14.99% under identical noise. These simulator values are for the 21-gate ansatz; the circuit submitted to live hardware is a distinct, more aggressively pruned instance (19 two-qubit gates, depth 76) whose noiseless one-way fidelity is 0.5662 (§6.5)—the two numbers describe different circuits, not conflicting reports of the same one. To

make the relationship between them precise: the submitted instance is *not* a separately re-optimized seed. It is the converged endpoint of the same per-pair optimization that yields the simulator ansatz, and its parameter trajectory self-prunes within the run—the intensity weights for one entangling pair collapse to ρ -sum $\sim 4 \times 10^{-4}$ (deleted by the polar-projection gradient, §5.2) and two further pairs reduce to single dominant Pauli terms, leaving 19 of the 21 two-qubit gates after transpilation onto the heavy-hex path. No pairs are excised post hoc, and no independent 19-gate optimization was performed; the 42-parameter vector that produced the hardware result is the same dual-MMA output recorded in the per-run JSONs. The lower noiseless fidelity of the submitted instance is therefore a self-pruning consequence: pruning reduces depth at some cost to ideal representation, and here that cost is paid on the noisy objective, whereas the 0.9829 is the ideal (noise-free) evaluation of the full 21-gate ansatz before the depth-for-fidelity trade is made. LPTE is about twice as resilient to noise as the exact unitary (Fig. 4; product-PTE floor $F = 0.6329$). To quantify robustness across noise environments, parameters optimized under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The transfer efficiency

$$\eta = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{\bar{F}_{ij}}{\bar{F}_{ii}}, \quad (7)$$

is the mean ratio of off-diagonal (cross-model) fidelity $\bar{F}_{ij}$ to diagonal (on-model) fidelity $\bar{F}_{ii}$, where $\bar{F}_{ij}$ is the mean DM fidelity over 10 seeds of parameters trained under model i and evaluated under model j ; diagonal entries ($i = j$) are excluded to avoid self-counting. The result is $\eta = 0.9943$ (Fig. 5): parameters trained under one model retain 99.4% of their on-model fidelity under any other. The scope of this transfer claim is qualified: it is demonstrated within the same parametrized noise-model family (the six synthetic variants of §4.2), not across categorically different noise channels such as correlated two-qubit faults or non-Markovian dynamics, whose cross-family transfer is untested here. Within that family, the structural well is not an overfit to any specific noise model.

6.5 Real hardware on ibm_fez

Lemma 3 (Lipschitz equivalence gap). *This is a standard Lipschitz-type bound of the fidelity under a compiled-gate gap, stated here as a lemma for convenient reference rather than as a new result. The optimizer evaluates the polar-projected U_{safe} , while hardware executes the compiled Trotter product U_{circ} with $\|U_{\text{safe}} - U_{\text{circ}}\| \leq \epsilon$ [16]. By Fuchs–van de Graaf [37], $D \leq \sqrt{1 - F} \leq \sqrt{c} \epsilon$, so*

$$|L_{\text{ideal}} - L_{\text{hardware}}| \leq K\sqrt{c} \epsilon. \quad (8)$$

Because ϵ is small by design, Eq. (8) bounds how far the hardware landscape can drift from the simulator landscape.

The optimized (ρ, τ) from FakeFez are submitted to live ibm_fez (Heron r2, 156 qubits), where the experiment itself

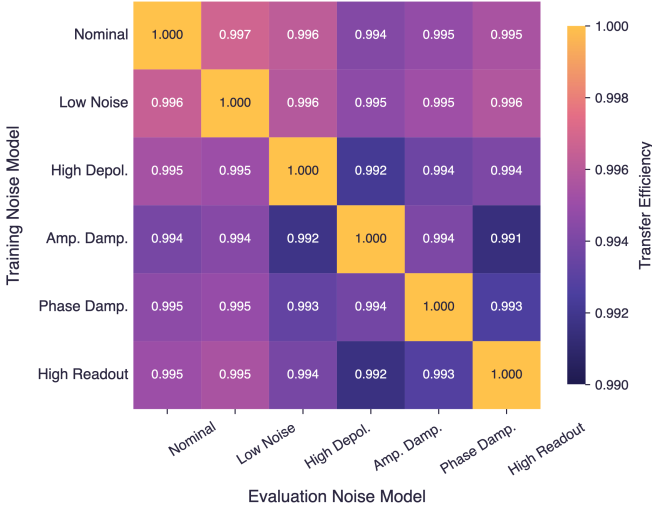


FIG. 5 Noise-model transfer matrix: DM fidelity of parameters trained under model i (rows) and evaluated under model j (columns), averaged over 10 seeds. Mean off-diagonal transfer efficiency $\eta = 0.9943$.

runs on the 8-qubit quietest path. State preservation is measured with a target-echo (inversion test) protocol: prepare the LPTE state with the optimized ansatz, then apply $U_{\text{target}}^\dagger$ and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of $U_{\text{target}}^\dagger$, so it is a strictly harder experiment than single-pass preparation. All hardware runs use `opt_level = 3` transpilation (the same level as all simulated results; see Table VII) on path [140, 141, 142, 143, 144, 145, 146, 147], the quietest path at measurement time. The optimized ansatz uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline (the target itself compiles to 21 gates at depth 82) and 14 gates at depth 56 for the MMA-optimized baseline. Two product-Trotter baselines are measured on the same day for a clean comparison: an *untuned* one with random gate parameters (the natural baseline, matching a naive product-Trotter submission) and an *optimized* one using the same dual-MMA routine TOPAZ applies to its own ansatz—isolating topology from parameter optimization. Both optimizations (LPTE and the MMA-optimized product baseline) ran the same dual-MMA solver on the same noise-aware objective with the same 30-iteration budget (recorded in Table VII and the per-run JSONs), so the baselines are matched in optimizer and budget, not merely in name. The reader may ask how a circuit of essentially the same size as the target (21 gates, depth 82) can reach only 0.1768 noiseless fidelity. The answer is not an expressibility ceiling: the product-Trotter ansatz is itself a product of the same $M = 21$ nearest-neighbour Pauli rotations used to define the target, so it can in principle represent it to arbitrarily high fidelity. The 0.1768 is instead the value the dual-MMA solver reaches on the *product-form* landscape under the matched budget—the same budget on which the per-pair (LPTE) form reaches 0.5662—so the gap is an optimization/landscape gap rather than a representational

one. This is exactly the claim the paper advances: LPTE’s relaxed per-pair form is easier for the optimizer to search, and the hardware gain is a co-design consequence of topology and search, not an artifact of an under-converged baseline. The simulator product-PTE value 0.6329 (Table VI) is not in conflict with the hardware 0.1768: those are different circuits (simulator product-PTE is 8 gates at depth 22; the hardware product baseline is 14 gates at depth 56 after transpilation onto the heavy-hex device), evaluated under different measures (DM fidelity under FakeFey noise versus noiseless one-way fidelity), so a reader should not expect them to coincide.

TABLE VII Live `ibm_fez` results, three same-day replications on the quietest path (2026-09-07, path [140, . . . , 147], `opt_level = 3`, $n = 2000$ shots per measurement). Measurement SEMs are binomial ($\sqrt{F(1-F)/n}$); the TOPAZ echo is reported as mean $\pm$ standard deviation across the three sessions (individual values in §6.6.5). Computed (noiseless) entries are exact statevector/density-matrix evaluations of the submitted circuit and carry no shot noise.

Configuration	Type	Fidelity
Product-Trotter, untuned random params (two-way echo)	measured	0.0478 $\pm$ 0.0034
Product-Trotter, MMA-optimised (two-way echo)	measured	0.1008 $\pm$ 0.0048
TOPAZ LPTE (two-way echo, mean of 3 sessions)	measured	0.2883 $\pm$ 0.0394
TOPAZ LPTE (two-way echo, individual sessions)	measured	0.2443 $\pm$ 0.0068, 0.3008 $\pm$ 0.0073, 0.3200 $\pm$ 0.0074
ZNE on TOPAZ echo (per session)	measured	0.2058, 0.3075, 0.3968
TPNM on TOPAZ echo (per session)	measured	0.2919, 0.3159, 0.3200
TOPAZ LPTE (one-way)	noiseless	0.5662
Product-Trotter, MMA-optimised (one-way)	noiseless	0.1768
Product-Trotter, untuned random (one-way)	noiseless	0.0758

To disambiguate the quantities reported here: the top seven rows are device measurements ($n = 2000$ shots each). $F_{\text{echo}} = 0.2883 \pm 0.0394$ is the mean measured target-echo—the return probability after $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$ on device, a protocol that doubles the forward gate count plus the transpilation overhead of $U_{\text{target}}^\dagger$. The three sessions (0.2443, 0.3008, 0.3200) span a single overnight window on the same quietest path, and their spread is the same order as the binomial SEM—the measurement is stable across replications. The one-way entries are not device measurements but the fidelity of the *hardware-submitted circuit* (19 two-qubit gates, depth 76) against the ideal target, evaluated exactly without shot noise: $F_{1\text{-way}} = 0.5662$ is the noiseless one-way state-preparation fidelity $|\langle \psi_{\text{target}} | \psi_{\text{circ}} \rangle|^2$. Because these are computed quantities they are reported without error bars. They are not interchangeable with the echo: the echo is a lower bound on the noisy one-way fidelity, not equal to it. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along the path.

6.5.1 The inversion-test echo and the role of ZNE

The hardware experiment measures a *target-echo* (inversion test): the return probability of $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$, which quantifies how closely the optimized state matches the target. Unlike a self-echo $U \cdot U^\dagger$ (which is trivially 1 for any unitary), the target-echo measures state-preparation fidelity directly through the round-trip protocol, so $F_{\text{echo}} = 0.2883 \pm 0.0394$ is a lower bound on the noisy one-way fidelity ($F_{1\text{-way}} = 0.5662$). The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along

the path.

Relative to the same-day, same-path untuned product-Trotter baseline (0.0478 ± 0.0034), the mean TOPAZ raw echo is a $6.04\times$ gain ($z \approx 22$, $p < 10^{-7}$; Fig. 6); against the MMA-optimised product baseline (0.1008 ± 0.0048) it is a $2.86\times$ gain ($z \approx 17$). Neither comparison is confounded by unequal representational capability: against the *same* target, the untuned product baseline’s noiseless one-way fidelity is 0.0758 and the optimised baseline’s is 0.1768, so TOPAZ represents the target $7.47\times$ better than the untuned random-parameter circuit and $3.20\times$ better than the optimised one under noise-free evaluation (0.5662 vs. 0.0758/0.1768). The raw-echo gain therefore tracks the noiseless representation gain, which the retention fractions confirm: under free evaluation both circuits retain comparable shares of their ideal fidelity through the noisy round-trip (50.9% for TOPAZ vs. 63.1% and 57.0% for the two baselines), so the $6.04\times$ raw-echo improvement is attributable to representation—not to any hidden advantage of a shallower circuit, since all three circuits are compared against the same inversion protocol.

Standard ZNE is *inconsistent* on this protocol: across the three sessions the extrapolated echo ranges from below the raw echo (0.2058 vs. 0.2443) to above it (0.3968 vs. 0.3200), and the mean never approaches the noiseless bound. Tensor-product noise-model (TPNM) correction—a noise-aware estimate that models the round-trip channel as a tensor product of per-gate depolarizing channels—behaves more stably (0.2919, 0.3159, 0.3200), tracking the raw echo across sessions. TOPAZ delivers its gain before noise accumulates, on the circuit itself, so the pre-circuit optimization is complementary to—and, on this protocol, more reliable than—extrapolation-based post-processing.

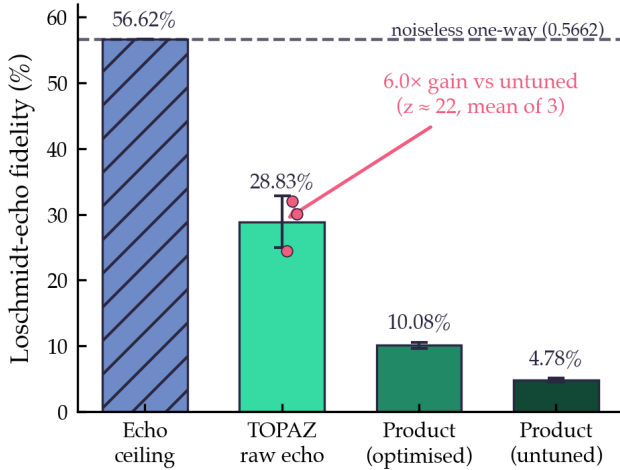


FIG. 6 Live `ibm_fez` target-echo (inversion test): TOPAZ raw echo versus product-Trotter baseline and standard ZNE.

6.6 Stacking with post-processing QEM

TOPAZ composes with post-processing because the two act on different parts of the pipeline: TOPAZ pre-tunes the parameter space, while ZNE, CDR, and VD correct observables. ZNE alone at the baseline gives marginal improvement ($0.0677 \rightarrow$

0.0747); on top of TOPAZ it adds a further 0.0667 DM gain. The operating point moved by TOPAZ lies on a part of the noise curve where extrapolation is more reliable. As in Table VI, ZNE, CDR, and VD are each applied to the LPTE circuit individually (our own implementations, run against the same simulated noise model); we did not run them as a combined stack, so the Table VIII rows are single methods rather than a full mitigation stack.

TABLE VIII 8-qubit mitigation stacking on FakeFaz (LPTE baseline). DM: density-matrix fidelity under the target noise model; Inv.: inversion-test fidelity (target-echo).

Configuration	DM Fidelity	Echo Fidelity
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9085	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

Each mitigation is applied individually to the LPTE circuit; a combined ZNE+CDR+VD

stack was not run, so the rows are not additive. “TOPAZ LPTE alone” is the same best-seed 21-gate circuit as Table VI (DM 0.9085, echo 0.7715).

6.7 Scaling to 16 qubits

To probe the representability ceiling of the LPTE ansatz beyond 8 qubits, we ran the full TOPAZ pipeline on `ibm_fez` (Heron r2) for a 16-qubit nearest-neighbour chain [112, 113, 131, 132, 133, 114, 115, 129, 130, 117, 116, 145, 146, 109, 110, 111] using the *per-pair* mode (90 parameters). The optimizer improves the return probability from 0.0003 (random) to 0.1151—a $383\times$ gain from the classical search alone—but this noiseless ceiling (11.5%) is far below the 56.6% achieved on the 8-qubit path, confirming that the LPTE ansatz’s expressibility collapses as the chain lengthens and the linear topology becomes increasingly incompatible with the heavy-hex device connectivity.

The unmitigated target-echo (raw echo) is 0.0065; TPNM correction gives 0.0103; our in-house ZNE (standard unitary folding, SF = 1, 3, single batched job) gives 0.0120. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.0065/0.1151 = 5.6\%$, well below the $\sim 51\%$ observed at 8 qubits. A separate ZNE-only run using the standard IBM unitary-folding protocol (SF = 1, 3, 5, Richardson extrapolation) yielded folds $0.00475/0.00625/0.0030$ —non-monotonic, the same inconsistency signature seen at 8 qubits (where the extrapolated echo ranged from below to above the raw echo across sessions). The Richardson fit extrapolated to 0.001497, which when clipped to the noiseless bound gives 0.002219, collapsing below the raw echo. ZNE’s instability on these deep round-trip circuits is therefore structural (intrinsic to deep round-trip circuits) rather than an implementation artifact: gate stretching pushes the circuit beyond the coherence window, and extrapolation frequently produces a result at or below the unmitigated measurement.

These 16-qubit results serve as a scaling-limitation data point: the LPTE ansatz’s representability ceiling (11.5% noiseless

fidelity) and the ZNE failure mode both worsen with qubit count, consistent with the deep-circuit decoherence picture. Extending the framework to larger systems would require adaptive pruning thresholds and device-side optimization—directions we discuss in §8.

7. Well geometry and sparsity

Across every seed, session, and backend (triple-channel, FakeFez, FakeMontrealV2, live ibm_fez), the trajectory shows the same shape: a rapid descent over the first 5–10 iterations, then a stable floor. The well is not a three-dimensional attractor but a family of sparse subspaces: each seed prunes to a different 8–24 dimensional subset of the 42-dimensional (ρ, τ) space, and the survivors are not the target Hamiltonian’s largest coefficients—the top six target terms are all pruned. The optimizer selects for noise robustness, not target matching. Figure 7 makes the family-of-wells picture concrete: no two seeds prune the same set of pairs.

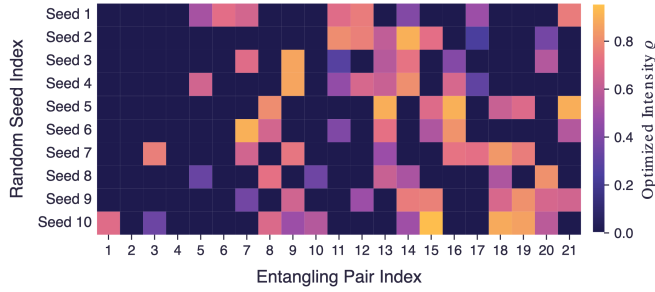


FIG. 7 Family-of-wells sparsity: final pair intensities $\kappa(p) = \sum_P r_{p,P}$ for 10 random seeds (rows) across the 21 LPTE entangling pairs (columns). Dark cells are pruned parameters ($\rho < 0.05$); each seed retains a different sparse subset.

7.1 Canyon-river curvature

We use the term *canyon-river* to describe the heterogeneity pattern in the Hessian eigenvalues at the well: the intensity subspace is uniformly steep ($\kappa_\rho \approx 1$, all ρ equally stiff at ~ 870), while the timing subspace has a few extremely stiff directions amid many soft ones ($\kappa_\tau \approx 9 \times 10^3$, timings vary by 8,000 $\times$ among survivors), with combined $\kappa \approx 2 \times 10^5$. The eigenvalue clustering into two sharp bands is consistent with dampened barren-plateau behaviour [17], but no eigenvalue-versus-qubit-count diagnostic was performed, so we make no claim of barren-plateau avoidance [18].

8. Discussion

8.1 Scope and limitations

All headline circuits are 8 qubits on a linear chain, and the sparsification means are over 5 seeds (10 for the ablation). The pruning threshold $\tau_\kappa = 0.05$ is a heuristic choice; systematic threshold sweeps across molecule sizes and noise levels remain future work. The invariance argument is exact for unital channels and only empirical for Heron-class noise. Hardware

experiments were conducted on a single day; reproducibility across different calibration sessions and hardware revisions is not yet established. A preliminary 16-qubit run (§6.7) confirmed the representability ceiling (11.5% noiseless fidelity vs. 56.6% at 8 qubits) and the same standard-ZNE instability (extrapolation collapsing below the raw echo), indicating that the 8-qubit headline results do not straightforwardly extend to longer chains without adaptive pruning and device-side optimization. Global PTE is computationally expensive for $N \geq 4$ and was not used for the headline experiments; a from-scratch TOPAZ run on critical-point TFIM still leaves a 0.4–1.2 Ha residual. Transfer across physical layouts (e.g. from a linear chain to a heavy-hex topology) was not explored; the current demonstration tests noise-model transfer only. The density-matrix engine makes simulation-path scaling intractable beyond ~ 20 qubits; the transfer efficiency $\eta = 0.9943$ mitigates this by letting offline-trained parameters move to hardware without re-optimization. The open problem is a device-side MMA iterate for cases where the simulator diverges from hardware noise at scale.

8.2 Relation to post-processing QEM

ZNE, CDR, and virtual distillation act on measurement statistics after the circuit runs; TOPAZ acts on the topology before execution. The two are complementary rather than competitive. Empirically, TOPAZ circuits preserve the noise-energy relationship that extrapolation methods rely on, so stacking TOPAZ with a post-processor improves that method’s reliability rather than merely adding a constant.

8.3 Future work

Several directions extend the current framework. (i) Device-side optimization: replacing the simulator-based dual-MMA with a hardware-in-the-loop iterate would eliminate the simulator-hardware gap and enable scaling beyond the ~ 20 -qubit simulation ceiling. (ii) Adaptive pruning: replacing the fixed threshold $\tau_\kappa = 0.05$ with a schedule that adapts to the loss landscape curvature could improve gate reduction on larger instances. (iii) Non-linear topologies: extending LPTE beyond linear chains to heavy-hex or modular architectures would match the connectivity of production hardware. (iv) Larger molecules: applying TOPAZ to systems beyond H_2/H_4 —such as LiH , BeH_2 , or small active-space problems—would test whether the structural-well phenomenon persists at greater chemical complexity. (v) Combined mitigation stacks: running ZNE+CDR+VD simultaneously (rather than individually) on TOPAZ circuits would quantify whether the methods’ gains are additive, sub-additive, or synergistic.

9. Conclusion

We have presented TOPAZ, a classical framework that optimizes quantum circuit topology by treating entangling-edge intensities as continuous design variables. The key insight is that the optimizer consistently converges to structural wells—basins in the noise-deformed parameter landscape, empirically characterized here, where the output is intrinsically robust.

On the 8-qubit FakeFey simulator, TOPAZ’s pruning step achieves a 34% depth reduction relative to its own unpruned ansatz at negligible fidelity cost; on the 156-qubit `ibm_fez` processor, the optimized circuit attains a 2.9× improvement in target-echo over the MMA-optimized product-Trotter baseline under matched parameter optimization, and 6.0× over the untuned baseline (mean of three same-day sessions, $z \approx 22$). Noise-free evaluation of the identical submitted circuits isolates the origin of the hardware gain: TOPAZ stores 3.2× more of the ideal target fidelity than the MMA-optimized baseline and 7.5× more than the untuned one, so on hardware the advantage is representational, with a 50.9% retention fraction of the one-way fidelity. The structural well transfers across synthetic noise models ($\eta = 0.9943$), and in simulation TOPAZ provides noise resilience complementary to post-processing methods like ZNE and CDR, whose reliability the shallow LPTE topology preserves under the noisy VQE benchmark. A preliminary 16-qubit experiment confirmed both the representability ceiling (11.5% noiseless fidelity, down from 56.6% at 8 qubits) and the same standard-ZNE instability, framing the 8-qubit headline results as a proof of concept whose extension to larger systems remains future work. The framework is limited to 8-qubit linear chains and single-day hardware demonstrations; nevertheless, the results establish that circuit topology—not just gate parameters—is a design variable worth optimizing.

Acknowledgements

We thank Sreekuttan L S, Nikhil Londhe, and Grishma Prasad at Bloq Quantum for their technical critiques and feedback across multiple iterations of this work. We also thank the faculty at The Shishukunj International School, Jhalaria for their early encouragement and support.

Author contributions

K.K. conceived the study, developed the TOPAZ framework and the parametric Trotter expression family, designed and ran the simulations and hardware experiments, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Data availability

All code, data, and analysis scripts for this work are available at the project’s public repositories (TOPAZ [20]; the companion continuous-topology manuscript [19]; the parametric-Trotter derivation [12]). The TOPAZ release used for the results of this manuscript is tagged v1.0, archived with its exact commit hash and full dependency environment in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449). Qiskit ≥ 0.46 and Qiskit Aer ≥ 0.14 were used; the pinned versions are recorded in `environment.yml`. Table-to-script mappings: Table V data is in `python/0-FINAL/results_h2_vqe/` and `python/0-FINAL/results_h4_vqe/` (per-seed logs in

`FINAL_PYTHON/rerun_logs/`); Table VI and Table II are produced by `benchmark_lpte.py`; Table I by `ablation.py`; Table VII by `run_ibmfez.py`; the 16-qubit data in `results_16q_hw/`; Figs. 4, 1, 2, 5, 6, 7 by the corresponding `plot_*` scripts in the same tree.

Hardware job records (Qiskit Runtime job IDs and per-circuit shot counts for each row of Table VII, together with the backend calibration timestamps captured at submission) are stored in `data/ibmfez_runs.jsonl`. These records, including the unredacted job IDs and the exact v1.0 release commit hash, are archived in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449) at publication time, so that every circuit measurement in Table VII is reproducible from the pinned code revision and raw runtime records.

Code availability

The optimization engine, Kraus contraction routines, and experiment drivers are released as open-source Python under the TOPAZ repository (DOI 10.5281/zenodo.19214449, tag v1.0).

Additional information

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